

Supplemental Material for: Non-additive Strain–Dopant Energetics in Quasi-Hexagonal C₆₀ Graphullerene

Xingqiang Chen and Qixing Wang
Department of Physics, Xiamen University, Xiamen 361005, China

I. FINITE-DIFFERENCE DERIVATION OF THE FOUR-CORNER COUPLING ESTIMATOR

This appendix derives the four-corner coupling estimator $\mathcal{S}$ used in the main text [Eqs. (2)–(3)] from a Taylor expansion of total energy about the pristine reference, justifies the stencil choice against alternatives, and states symmetry, gauge, and error-propagation properties.

A. Taylor expansion and discrete composition

Let $E(\epsilon, \delta)$ denote the total energy of a supercell with biaxial strain parameter ϵ and composition label δ (pristine reference vs. one substitutional dopant per C₆₀ on the fixed production map). Although δ is sampled on a discrete chemical grid (B, N, or P) rather than a continuous concentration field, the expansion below is the standard *finite-difference* representation of the underlying composition-dependent energy functional $E(\epsilon, \delta)$ on the audited (ϵ, δ) stencil—not a claim that dopant concentration is a differentiable continuum variable.

Expand about $(\epsilon_0, \delta_0) = (0, 0)$ to second order in increments $(\Delta\epsilon, \Delta\delta)$:

$$E(\epsilon_0 + \Delta\epsilon, \delta_0 + \Delta\delta) = E_0 + E_\epsilon \Delta\epsilon + E_\delta \Delta\delta + E_{\epsilon\delta} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2), \quad (1)$$

where $E_\epsilon \equiv \partial E / \partial \epsilon|_{0,0}$, $E_\delta \equiv \partial E / \partial \delta|_{0,0}$, and $E_{\epsilon\delta} \equiv \partial^2 E / \partial \epsilon \partial \delta|_{0,0}$. The bilinear coefficient $E_{\epsilon\delta}$ (denoted γ in continuum discussions) characterizes the intrinsic mixed energetic response; it is a *material functional* of the host, boundary conditions, and exchange–correlation approximation. Eq. (1) requires sufficiently small perturbations and a single, smooth energy branch; it fails across phase transitions or when strain drives large bond rearrangements (main text, Sec. IV F).

B. Four-corner stencil and cancellation of linear terms

Define the four corner totals on the rectangular grid $(\epsilon, \delta) \in \{(0, 0), (\Delta\epsilon, 0), (0, \Delta\delta), (\Delta\epsilon, \Delta\delta)\}$: E_{00} , $E_{\epsilon 0}$, $E_{0\delta}$, $E_{\epsilon\delta}$. The symmetric combination

$$\Delta E_{\text{cross}} \equiv E_{\epsilon\delta} - E_{\epsilon 0} - E_{0\delta} + E_{00} \quad (2)$$

substitutes Eq. (1) at each corner. All first-order terms cancel identically:

$$\Delta E_{\text{cross}} = E_{\epsilon\delta} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2). \quad (3)$$

Dividing by N_{atoms} gives the per-atom estimator $\mathcal{S} = \Delta E_{\text{cross}} / N_{\text{atoms}}$ [Eq. (3)], i.e.,

$$\mathcal{S} \approx \frac{E_{\epsilon\delta}}{N_{\text{atoms}}} \Delta\epsilon \Delta\delta + \mathcal{O}(\Delta\epsilon^2, \Delta\delta^2, \Delta\epsilon^2 \Delta\delta, \Delta\epsilon \Delta\delta^2), \quad (4)$$

which is Eq. (4) of the main text. Thus $\mathcal{S}$ *estimates* the mixed energetic response under finite $(\Delta\epsilon, \Delta\delta)$; it is not itself the continuum coefficient $\gamma \equiv E_{\epsilon\delta} / N_{\text{atoms}}$ unless the higher-order remainder is negligible.

The additive-screening bias is, by construction, $\mathcal{S} = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)]$ [Eq. (2)], where each ΔE is measured relative to the same pristine reference $E(0, 0)$.

C. Why this four-point stencil?

The symmetric four-corner form is *not* unique, but it is the minimal second-order-accurate stencil on a rectangular (ϵ, δ) grid that (i) requires only four DFT totals, (ii) cancels all first-order strain and composition increments and any shared reference offset, and (iii) isolates the bilinear cross term in one algebraic step. Alternatives differ in cost, bias, or interpretability:

- **Central (five-point) stencils** on a finer grid reduce truncation error at the price of additional corners and, for discrete δ , may interpolate between non-physical composition labels.
- **Least-squares fits** of $E(\epsilon, \delta)$ to a low-order polynomial over many points can estimate $E_{\epsilon\delta}$ but mix truncation error with fit-window choice and do not map one-to-one onto the additive-screening bias audited here.
- **Full mixed Hessian** $\partial^2 E / \partial \epsilon \partial \delta$ from analytic or numerical derivatives of relaxed energies requires continuous, well-defined δ and is ill posed on the present discrete substitution grid.

Asymmetric three-point schemes retain linear contamination in at least one channel; Table II contrasts $\mathcal{S}$ with other descriptors.

D. Symmetry, gauge invariance, and units

a. Symmetry (Clairaut). If $E_{\epsilon\delta} = E_{\delta\epsilon}$ (Schwarz/Clairaut theorem for a smooth E), the symmetric stencil (2) is invariant under exchanging the

TABLE I. Positioning relative to representative prior work on qHP graphullerene and related covalent networks (not an exhaustive literature map). “Joint (ϵ, δ) ” denotes simultaneous strain and substitutional doping in the same calculation series; “four-corner total-energy audit” denotes symmetric (ϵ, δ) corner differences as in the main-text definition of $\mathcal{S}$.

Prior focus	Joint (ϵ, δ)	Four-corner E	Nonlinear audit
Separate strain or doping maps [1–5]	×	×	×
Nonlinear band-edge / transport response [6–8]	Partial	×	×
Alloy cluster expansion (effective $c\epsilon x$) [9]	✓	×	×
This work (qHP C ₆₀ , PBE+D3)	✓	×	×

order in which the ϵ and δ increments are applied on the grid: the estimated cross term is the same whether one views strain or composition as the “first” axis. This is a property of the *stencil*, not a claim that biaxial strain and chemical substitution are physically interchangeable controls (main text, Sec. II C).

b. Gauge invariance. Adding an arbitrary constant C to all four corner totals leaves $\mathcal{S}$ unchanged; only energy *differences* among the four corners enter. Shared systematic shifts (e.g., uniform XC offsets) therefore cancel in the four-corner combination when all corners share the same computational protocol.

c. Units. $\mathcal{S}$ is reported in meV per atom. The dimensionless additive-failure fraction $\eta = |\mathcal{S}|/|\Delta E_{\text{coupled}}|$ [Eq. (6)] has no units. The continuum coefficient $\gamma \equiv \partial^2 E / \partial \epsilon \partial \delta$ carries units of energy per atom per strain increment per composition increment; on the production grid ($\Delta\epsilon = 3\%$, discrete $\Delta\delta$ from pristine to one substituent per C₆₀), $\mathcal{S} \approx \gamma \Delta\epsilon \Delta\delta$ up to higher-order remainders.

E. Error propagation

With $\mathcal{S} = (E_{\epsilon\delta} - E_{\epsilon 0} - E_{0\delta} + E_{00})/N_{\text{atoms}}$ and independent, equal per-atom uncertainties σ_e on each converged corner total,

$$\sigma_{\mathcal{S}}^2 = \frac{1}{N_{\text{atoms}}^2} (\sigma_e^2 + \sigma_e^2 + \sigma_e^2 + \sigma_e^2) = 4\sigma_e^2, \quad \sigma_{\mathcal{S}} = 2\sigma_e, \quad (5)$$

which is Eq. (5) of the main text when corners are uncorrelated. Correlated SCF noise across chemically related corners would reduce the effective coefficient below 2; the main-text reporting band uses $2\sigma_e$ as a conservative upper bound (Table I).

II. LITERATURE POSITIONING AND DESCRIPTOR COMPARISON

III. ARCHIVED LEGACY-PBE TETRAMER BASELINE

Table III reproduces the archived rigid-strain grid (PBE without D3) retained for reproducibility; it is *not*

used for quantitative claims in the main text.

IV. CALCULATION INVENTORY (REPRODUCIBILITY)

Tables IV and V list the DFT groups underlying the main-text claims and superseded exploratory grids retained only for reproducibility. They are *not* scientific results tables.

V. FULL TETRAMER π -DOS AT $\epsilon=0$

VI. PERIODIC FRONTIER-ORBITAL LOCALIZATION

Cage-resolved VBM/CBM isosurfaces on periodic $n \times \text{C}_{60}$ cells ($n = 2-8$) at $\epsilon=0$ complement the tetramer π -DOS above and the main-text electronic taxonomy (donor/acceptor vs. size dilution). Carbon cages are shown as uniform gray MSMS molecular surfaces (probe 1.3 Å); heteroatoms use element colors.

VII. PERIODIC COUPLING-ENERGY ROWS ($n \geq 6$)

VIII. FOUR-CORNER TOTAL ENERGIES (+3%)

IX. MARCUS REORGANIZATION ENERGIES

Polaron transport diagnostics are reported here for completeness; the main-text analysis centers on $\mathcal{S}$.

X. HOPPING INTEGRAL J

On the tetramer dimer grid, B substitution at $\epsilon=0$ delocalizes the charged-state orbital (IPR 75.0 $\rightarrow$ 24.6) while raising J from 26.6 to 46.4 meV; coupled B @ +3% gives $J = 37.2$ meV and IPR 23.1. The criterion $J > \lambda/2$ for polaron hopping dominance [4] is *not* met at PBE+D3 ($J < \lambda/2$), so a polaron transition is not confirmed on this grid.

TABLE II. Positioning of the strain-dopant coupling energy $\mathcal{S}$ (main-text four-corner estimator) against common stability and electronic descriptors. Algebraically $\mathcal{S}$ has the form of an interaction (coupling) energy; the distinction is *dimensional*, i.e. which axes of the joint (ϵ, δ) control space each quantity resolves. “Additive-screening error” denotes the bias incurred when total energies under joint load are assembled additively from separate strain-only and doping-only scans; $\mathcal{S}$ equals that bias by construction. “Double-reference cross term” denotes a symmetric difference of differences over the four (ϵ, δ) corners rather than subtraction of a single reference state or fragment pair. Entries are conceptual scope, not numerical performance; consistent with its energetic definition, $|\mathcal{S}|$ is *not* reducible to a single geometric ($\Delta\bar{d}$) or charge (Δq) channel alone.

Descriptor	Nonlinear (ϵ, δ) coupling	Additive-screening error	Double-reference cross term	Joint strain + doping design
Formation / substitution energy E_{sub}	×	×	×	×
Binding / interaction energy ΔE_{int}	Partial	×	×	Partial
Linear strain coefficient α	×	×	×	Partial
Band gap E_g / work function	×	×	×	×
$\mathcal{S}$ (this work)	✓	✓	✓	✓

TABLE III. Archived legacy-PBE (no D3) tetramer rigid-strain baseline retained for reproducibility only; not used for main-text quantitative claims. $E_{\text{sub}} = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ at $\epsilon=0$ (no chemical potentials; relative ranking within this protocol only).

System	n_{dop}	$E(\epsilon=0)$ (E_h)	$E(\epsilon=+5\%)$ (E_h)	α (meV/%)	E_{sub} (eV/dopant)
Pristine	—	−1363.5512	−1363.5509	−1.0	—
B-doped	13	−1328.4900	−1328.4090	+56.9	+73.4
N-doped	12	−1414.3812	−1414.4053	−303.2	−115.3
P-doped	12	−1370.4965	−1370.4888	−19.1	−15.8

-
- [1] H. Li, M. A. Shakoori, and R. Wang, strain engineering of electronic structure and thermoelectric properties of quasi-hexagonal fullerene monolayer, *J. Appl. Phys.* **136**, 015101 (2024).
 - [2] M. A. Khan and M. N. Leuenberger, tuning the electronic and optical properties of impurity-engineered two-dimensional graphullerene half-semiconductors, *Phys. Rev. Mater.* **9**, 034001 (2025).
 - [3] B. Peng and M. Pizzochero, monolayer C_{60} networks: a first-principles perspective, *Chem. Commun.* **61**, 10287 (2025).
 - [4] A. Capobianco, A. Breglia, P. Gentile, G. Capano, A. Asinari, G. Drera, L. Sangaletti, M. Cossi, and S. Casassa, electron localization and mobility in monolayer fullerene networks, *Nano Lett.* **24**, 7891 (2024).
 - [5] R. M. Tromer, L. A. R. Junior, and D. S. Galvao, A DFT study of the electronic, optical, and mechanical properties of a recently synthesized monolayer fullerene network, *Chem. Phys. Lett.* **804**, 139925 (2022).
 - [6] S. Chen, H. Liu, and K. Wang, Untangling the respective effects of heteroatom doping and strain on electronic transport in 2D materials, *Adv. Mater.* **36**, 2301234 (2024).
 - [7] P. Silva, R. M. Faria, L. G. Ferreira, and J. L. F. Fatima, large polarons in two-dimensional fullerene networks: the crucial role of anisotropy in charge transport, *Nanoscale* **16**, 2337 (2024).
 - [8] P. Santra, S. Ghaderzadeh, M. Ghorbani-Asl, H.-P. Komsa, E. Besley, and A. V. Krashennnikov, strain-modulated defect engineering of two-dimensional materials, *npj 2D Materials and Applications* **8**, 33 (2024).
 - [9] A. van de Walle and V. Blum, Cluster expansions in alloy theory, *MRS Bull.* **34**, 681 (2009).

TABLE IV. DFT calculation groups supporting the main-text claims (reproducibility inventory; Supplemental Material only). Superseded exploratory grids appear in Table V.

Calculation group	System	Count	Protocol
Archived baseline (legacy PBE)	$4 \times \text{C}_{60}$	24	Rigid strain; not used for main-text claims
π -DOS and gaps	$4 \times \text{C}_{60}$	12	PBE+D3
IPR (polaron dimer)	Dimer	4	Pristine/B; $\epsilon=0, +3\%$
Hopping J	Dimer	4	PBE+D3
Charged polarons ^a	$4 \times \text{C}_{60}$	20	UKS; geometry opt. + vertical SP
Periodic size scaling	$n \times \text{C}_{60}$	41	PBE+D3; 400/350 Ry
Cutoff benchmark	$6 \times \text{C}_{60}$ N	1	+3%; 400 Ry single-point
Ionic relaxation	$4 \times \text{C}_{60}$	4	Fixed-cell GEO; P corners
Ionic relaxation	$1 \times \text{C}_{60}$ P	4	Fixed-cell GEO; periodic
Alternate placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Periodic placement	$4 \times \text{C}_{60}$	16	$n=4$; two independent maps
Matched-functional rigid	$4 \times \text{C}_{60}$	4	PBE+D3 at relaxed coordinates
Reference placement	$4 \times \text{C}_{60}$	24	Rigid PBE+D3
Hirshfeld population	$1 \times \text{C}_{60}$ B/N/P	18	Spin-polarized KS

^aSix relaxed tetramer geometries provide charged-state initial structures.

TABLE V. Exploratory DFT grids superseded by the production protocol (reproducibility archive only).

Task group	System	N_{conv}	Disposition
Early strain probe	$2 \times \text{C}_{60}$	5	Superseded
Single-molecule doping	C_{60}	11	Superseded
Early electronics	$2 \times \text{C}_{60}$	24	Superseded
Exploratory design grid	$4 \times \text{C}_{60}$	20	Archive only

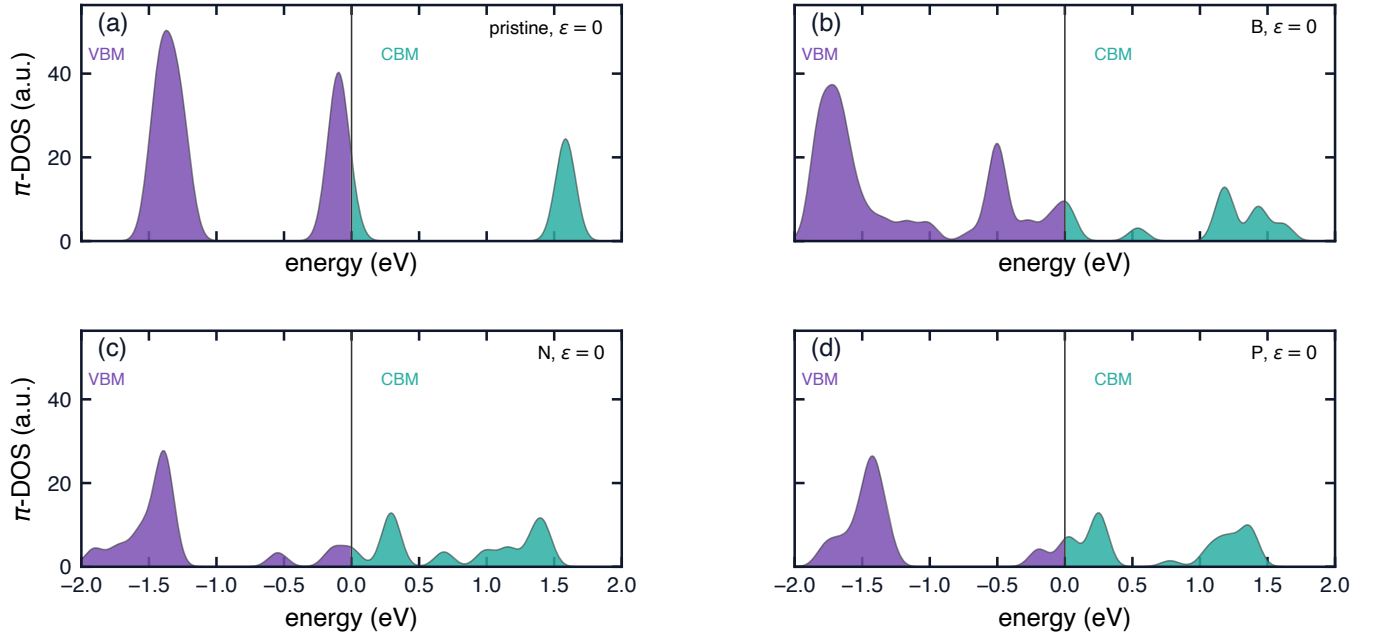
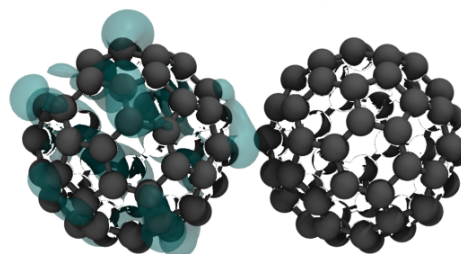


FIG. 1. Full π -DOS panels for pristine and B/N/P-doped tetramers at $\epsilon=0$ (PBE+D3), complementary to the main-text electronic-structure figure.

qHP C_{60} graphullerene ($\epsilon = 0\%$, N-doped)

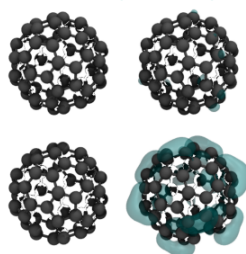
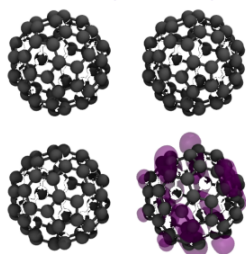
$n = 2$ ($2 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 2$ ($2 \times C_{60}$, N-doped)
CBM (LUMO)



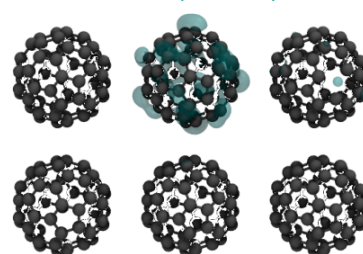
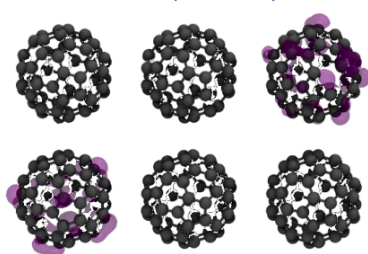
$n = 4$ ($4 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 4$ ($4 \times C_{60}$, N-doped)
CBM (LUMO)



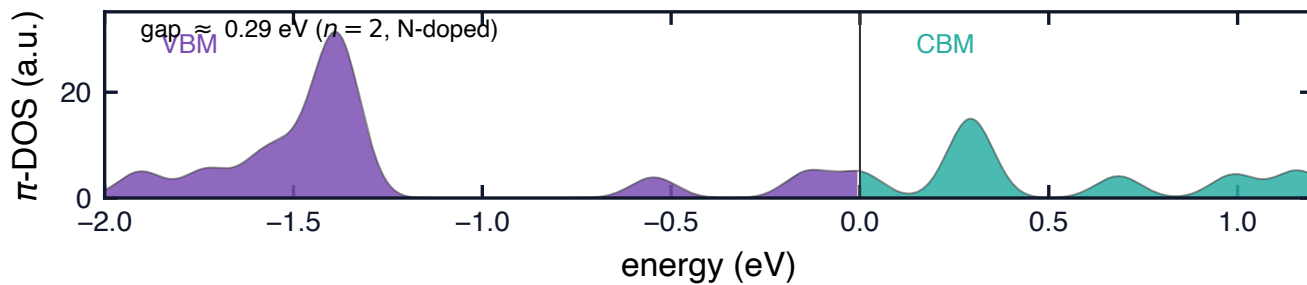
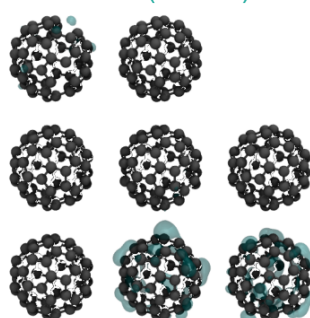
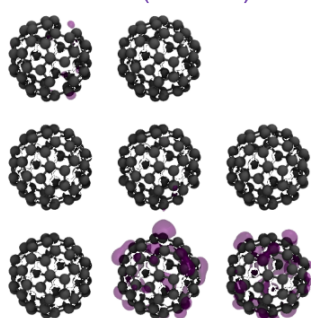
$n = 6$ ($6 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 6$ ($6 \times C_{60}$, N-doped)
CBM (LUMO)



$n = 8$ ($8 \times C_{60}$, N-doped)
VBM (HOMO)

$n = 8$ ($8 \times C_{60}$, N-doped)
CBM (LUMO)



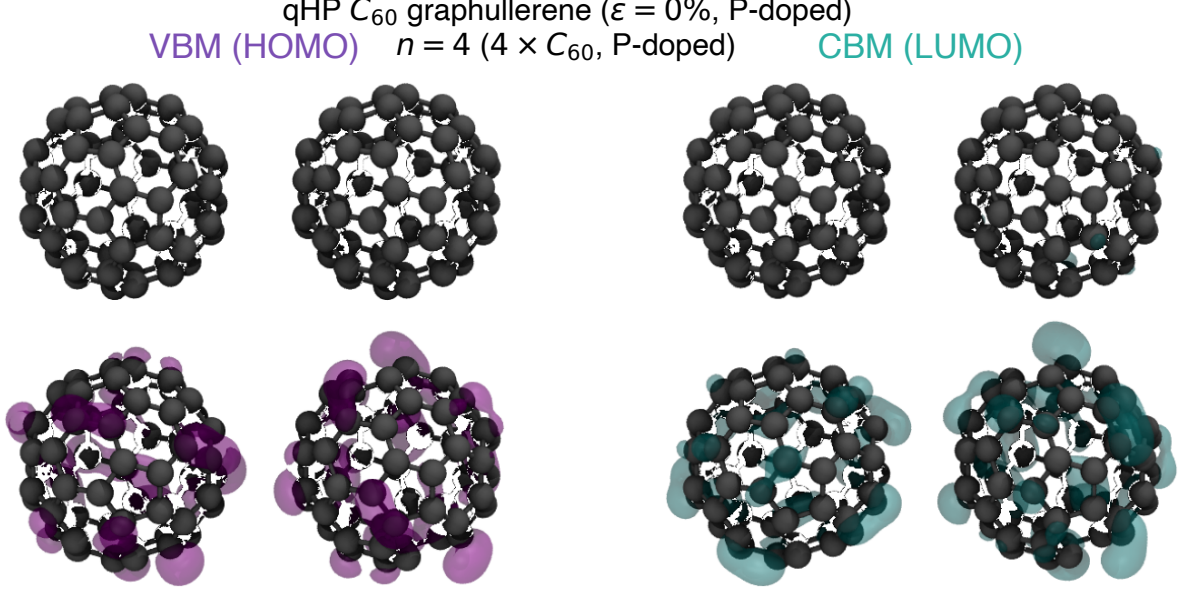


FIG. 3. P-doped $n=4$ supercell at $\epsilon=0$: VBM (left) and CBM (right) on gray MSMS cages. Both frontier states localize on a subset of cages, providing a spatial counterpart to the large periodic $|S|$ of size-mismatched P reported in the main text.

TABLE VI. Supplemental periodic coupling-energy rows at +3% ($n \geq 6$; 350 Ry production cutoff; completeness only).

n	Dopant	ϵ (%)	$\mathcal{S}$ (meV/atom)
6	B	+3	-7.76
6	N	+3	+2.38
6	P	+3	-31.89
8	B	+3	-11.26
8	N	+3	-2.30
8	P	+3	-29.94

TABLE VII. Machine-readable four-corner total energies per atom (Ha/atom) underlying $\mathcal{S}$ at +3% biaxial strain (PBE+D3, rigid protocol, reference placement seed 42). Corners follow Eq. (3): $E_{00} = E(0,0)$, $E_{e0} = E(\Delta\epsilon,0)$, $E_{0\delta} = E(0,\Delta\delta)$, $E_{e\delta} = E(\Delta\epsilon,\Delta\delta)$, all per atom relative to the same supercell size $n \times C_{60}$. $\mathcal{S}$ (meV/atom) matches Eq. (2) and the open audit chain (CSV mirror in the public data release).

n	δ	ϵ (%)	E_{00}	E_{e0}	$E_{0\delta}$	$E_{e\delta}$	$\mathcal{S}$
1	B	+3	-5.6826713370	-5.6825614149	-5.6339765778	-5.6341744697	-8.38
1	N	+3	-5.6826713370	-5.6825614149	-5.7532834847	-5.7530890722	+2.30
1	P	+3	-5.6826713370	-5.6825614149	-5.6922302079	-5.6932928189	-31.91
2	B	+3	-5.6817028549	-5.6817462745	-5.6338466634	-5.6339702803	-2.18
2	N	+3	-5.6817028549	-5.6817462745	-5.7532007583	-5.7529924491	+6.85
2	P	+3	-5.6817028549	-5.6817462745	-5.6895002016	-5.6905831259	-28.29
4	B	+3	-5.6828203886	-5.6827081746	-5.6341906516	-5.6343073761	-6.23
4	N	+3	-5.6828203886	-5.6827081746	-5.7535095848	-5.7532889091	+2.95
4	P	+3	-5.6828203886	-5.6827081746	-5.6924497414	-5.6932089910	-23.71
6	B	+3	-5.6828517956	-5.6827379495	-5.6340195050	-5.6341908867	-7.76
6	N	+3	-5.6828517956	-5.6827379495	-5.7533481235	-5.7531468231	+2.38
6	P	+3	-5.6828517956	-5.6827379495	-5.6916511984	-5.6927091482	-31.89
8	B	+3	-5.6828645675	-5.6825524079	-5.6341364256	-5.6342380770	-11.26
8	N	+3	-5.6828645675	-5.6825524079	-5.7534470201	-5.7532194756	-2.30
8	P	+3	-5.6828645675	-5.6825524079	-5.6919097127	-5.6926978180	-29.94

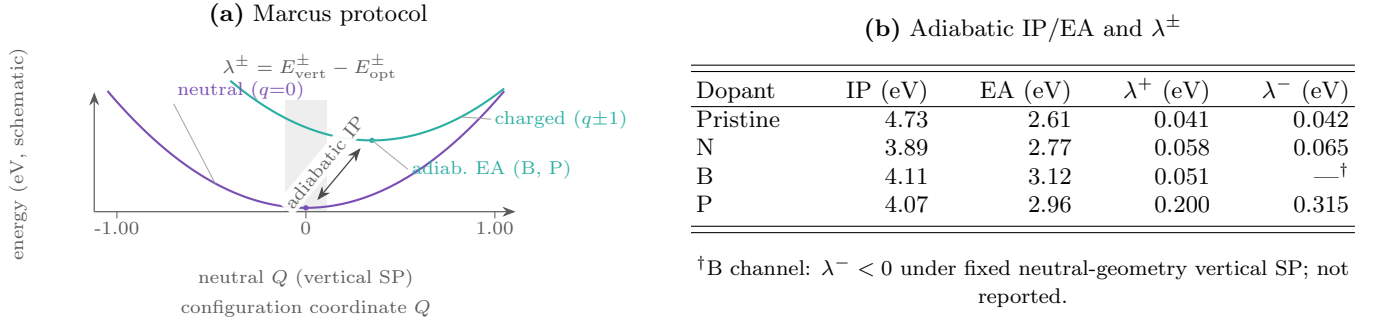


FIG. 4. Marcus reorganization energies from charged-state geometry optimizations and vertical single-points at the neutral geometry. Panel (a) schematically defines the vertical-adiabatic decomposition following Ref. 4; panel (b) tabulates adiabatic IP/EA and $\lambda^{\pm} = E_{\text{vert}}^{\pm} - E_{\text{opt}}^{\pm}$ for pristine and B/N/P tetramers. The electron-channel λ^{-} for B is unphysical under the fixed neutral-geometry vertical protocol and is excluded (Limitations).

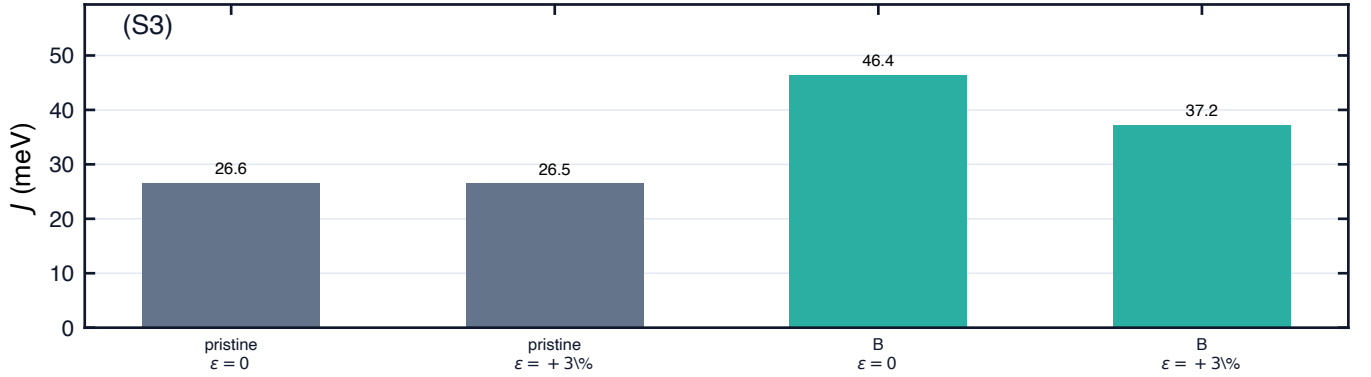


FIG. 5. Nearest-neighbor electronic coupling J on the tetramer dimer grid (pristine and B-doped cells at $\epsilon=0$ and $+3\%$; PBE+D3). Companion inverse-participation-ratio values for the same four corners are tabulated in the open data release (Methods).